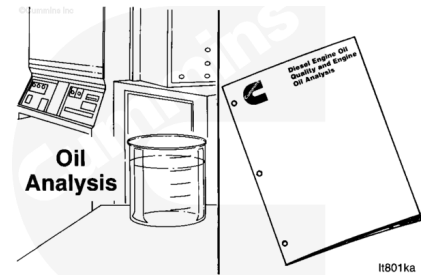


Trainings ▾

General Information

A used oil analysis can help diagnose internal damage and determine if it was caused by one of the following:

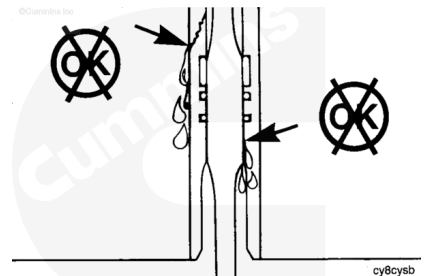
- Oil diluted with coolant
- Oil diluted with fuel



Pressure Test

Coolant-Diluted Lubricating Oil

For oil diluted with coolant; refer to Procedure .

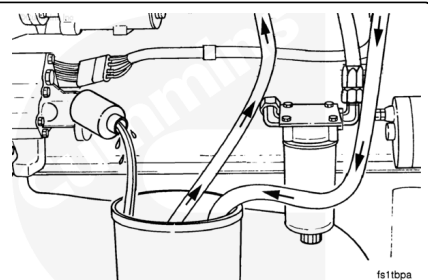


Fluorescent Dye Tracer

Lubricating Oil Dilution

Note : This test is **not** effective on a cold engine, less than 21°C [70°F] coolant temperature, or with a loose overhead setting.

The following is for lubricating oil diluted with fuel.



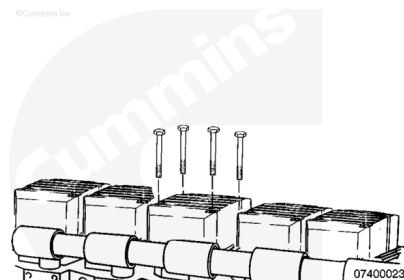
Install an isolated fuel supply tank to the inlet and drain lines.
Add fluorescent tracer, Part Number 3376891, or equivalent, to the fuel supply tank.

with Mechanically Actuated Injector

Remove the valve cover mounting capscrews, but do **not** remove the valve covers.

Refer to Procedure 003-011

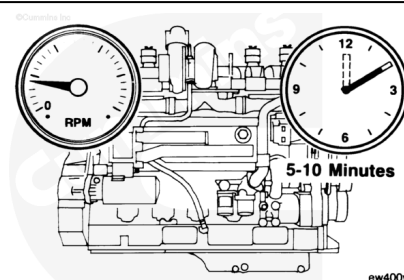
(/qs3/pubsys2/xml/en/procedures/20/20-003-011-tr.html).



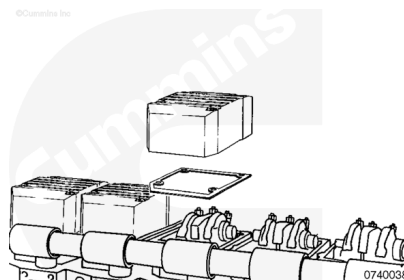
Start the engine and operate at high idle for 30 seconds.

Let the engine go to low idle.

Complete the next step within 5 minutes.

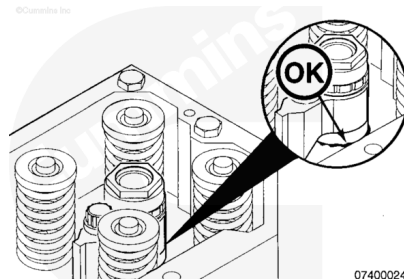


While the engine is operating at low idle, remove the valve covers. Refer to Procedure .



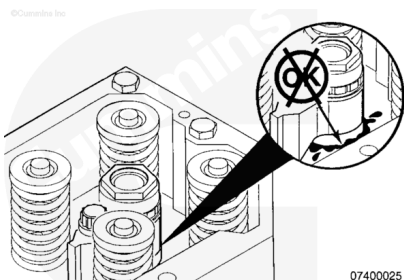
Note : Injectors normally have a small amount of fuel leakage. Fuel will be a yellow color.

Use a black light, Part Number 3163337, Part Number 3163338, or equivalent, to find fuel leaks from around the injector. Refer to the OEM manufacture for black light operating instructions.

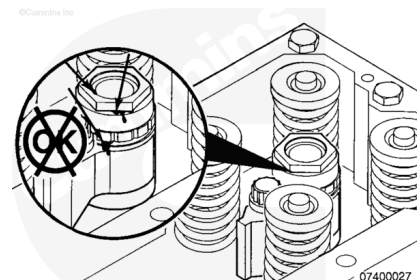


If there is excess leakage around the outside of the injector, check the top o-ring for damage.

Remove the injector and replace all three o-rings. Refer to Procedure .

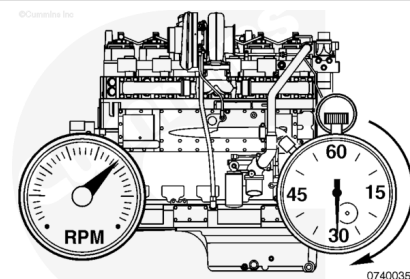


If there is excess leakage from inside the injector, replace the injector. Refer to Procedure .



with Electronically Actuated Injector

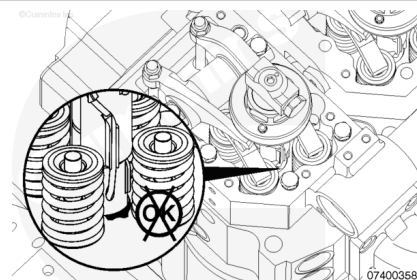
Start the engine and operate it at high idle for 30 seconds.



Remove the valve covers. Refer to Procedure .

Use a black light, Part Number 3163337, Part Number 3163338, or equivalent, to find fuel leaks from around the injector. Refer to the OEM for black light operating instructions.

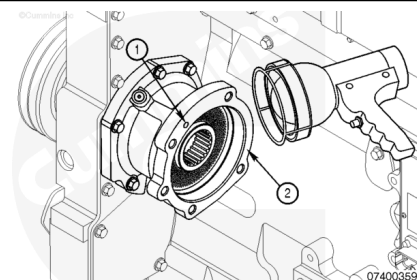
If there is excessive leakage around the injector, check to make sure the injector hold down clamp capscrew is **not** loose. Refer to Procedure for torque on the injector hold down capscrew.



If there is excessive leakage and the injector capscrew is properly tightened, check the injector o-rings for damage.

Remove the injector and replace the two o-rings. Refer to Procedure .

If leakage is **not** detected around any of the injectors, then remove the fuel pump. Refer to Procedure .



Note : The fuel pump is lubricated using engine lubrication. An internal fuel leak of the fuel pump can dilute the engine oil.

Use a black light to find leaks on the fuel pump and on the mating area (2) of the accessory drive where the oil passages (1) are located. If leakage is detected replace the fuel pump.

Refer to Procedure 005-016

(</qs3/pubsys2/xml/en/procedures/20/20-005-016-tr.html>).

Last Modified: 19-Feb-2007
